FEU INSTITUTE OF TECHNOLOGY

COLLEGE OF COMPUTER STUDIES AND MULTIMEDIA ARTS

CS0065
(INTELLIGENT SYSTEMS)

TECHNICAL ASSESSMENT

2

RULE – BASED AGENT

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Date:	30/09/2026

Github repo link: https://github.com/dmsison-ph/Sison-DonMarco-CS0065-AN42

I. BACKGROUND INFORMATION

A Rule-Based Agent is a type of intelligent agent that operates by following predefined rules to decide its actions. In this lab, students will work with a vacuum cleaner agent that can clean rooms or move between rooms depending on their state (dirty or clean).

The goal is to understand how agents perceive the environment, act according to rules, and update the environment.

II. TECHNICAL ASSESSMENT

RULE – BASED AGENT

Objective

At the end of this laboratory activity, students are expected to:

1. Explain the concept of rule-based agents.
2. Implement a rule-based agent in Python.
3. Interact with an environment by perceiving and acting based on rules.
4. Observe and analyze the behavior of an agent in a controlled environment.

Materials Needed

- Computer with Python 3.x installed
- Text editor or IDE (e.g., VSCode, PyCharm)
- Optional: Jupyter Notebook for interactive testing

General Instructions

1. This **Technical Assessment (TA4)** must be completed individually.
2. Follow the Task below.
3. Include clear comments in your code to describe its logic.
4. Each program should display the correct input, process, and output.
5. Ensure that your script displays clear and readable outputs.
6. Compile all Python codes and corresponding outputs into one PDF file.
7. Save it as **TA2_lastname.pdf**
   (Example: TA2_Patilano.pdf)
8. Submit the **python file** and **PDF** file.

Laboratory Task

Pre-Lab Preparation

1. Review the Python concepts:
   - Classes and objects
   - If-Else statements
   - Loops (for loop)

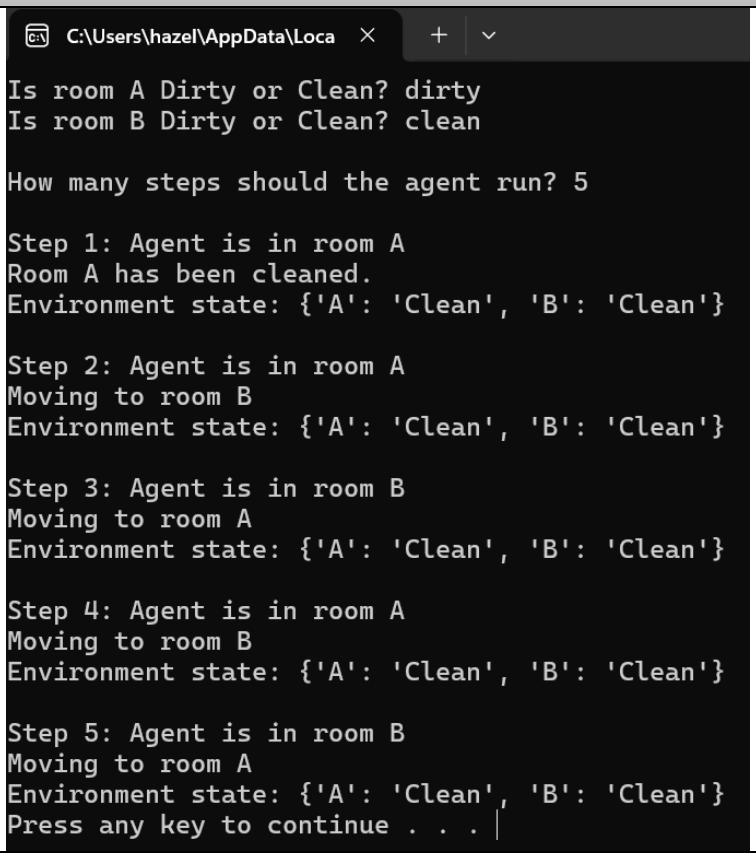
- Functions / methods

2. Understand the environment:
   - Two rooms: A and B
   - Each room can be Dirty or Clean

Task 1: Set Up the Environment

1. Initialize two rooms using a Python dictionary.
2. Allow the user to set the **initial state** of each room (dirty or clean).
3. Write a function to **check if a room is dirty** and a function to **clean a room**.

Sample Output



Task 2: Implement the Rule-Based Agent

1. Create a class for the agent.
2. Implement the **perceive-and-act method**:
   - Rule 1: If the current room is dirty, clean it.
   - Rule 2: If the current room is clean, move to the other room.
3. Implement a **move method** to switch rooms.

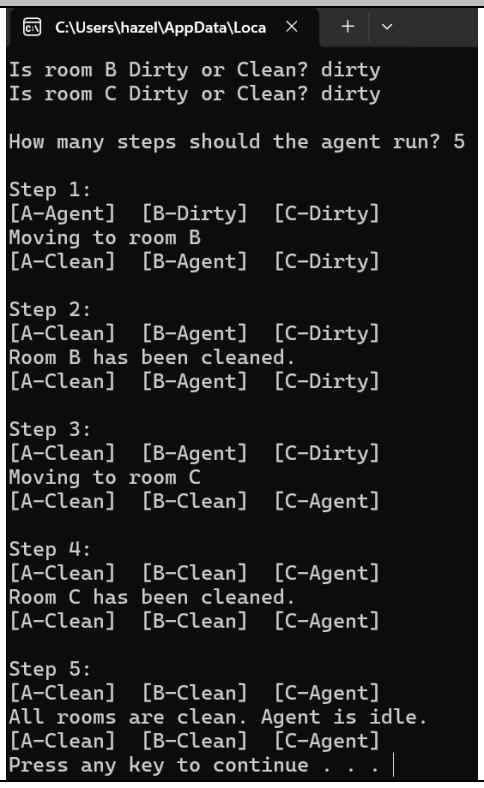
Task 3: Run the Simulation

1. Ask the user how many steps the simulation should run.
2. Run the agent in a loop for the specified steps.
3. After each step, print:
    - The agent's current location
    - The action taken (cleaned/moved)
    - Current state of the environment

Optional (Bonus Task)

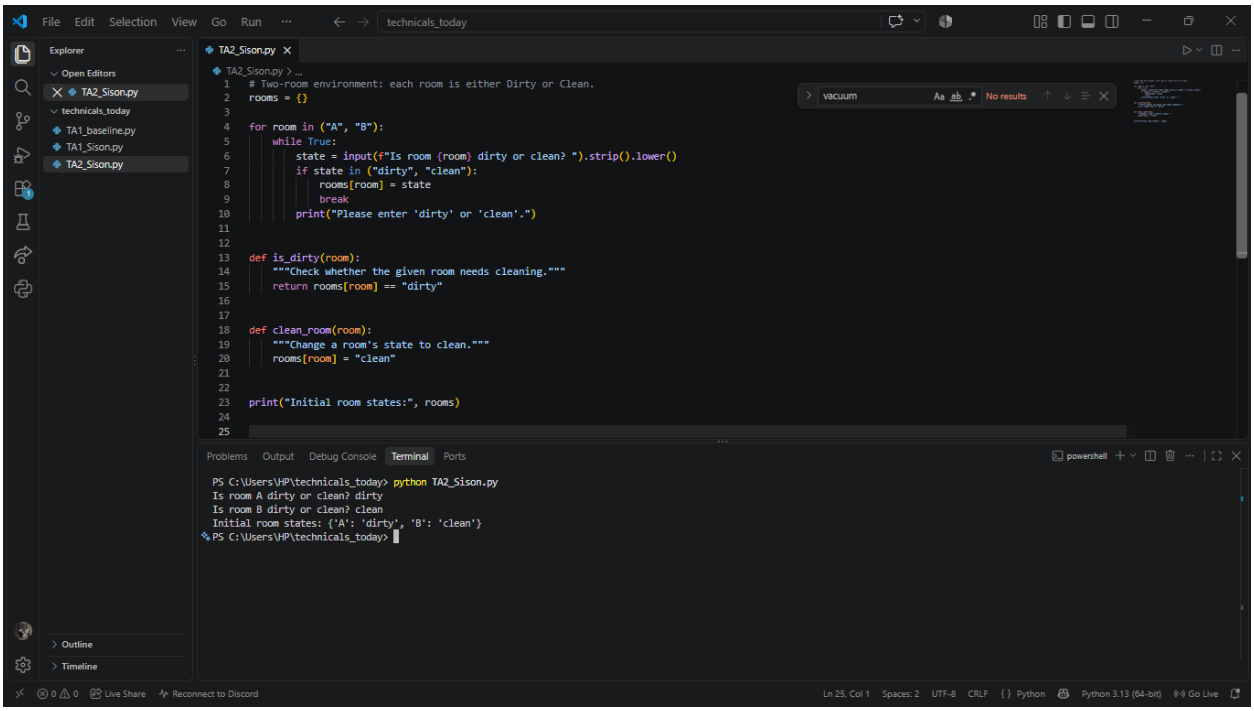
- Add a **third room (C)** and update the agent logic.
- Make the agent **randomly choose which dirty room to clean next**.
- Visualize the agent moving between rooms using a **text-based grid**.

Sample Output (Bonus Task)



Your Code - Laboratory Tasks ⇩

Task 1:



Code:

```
rooms = {}

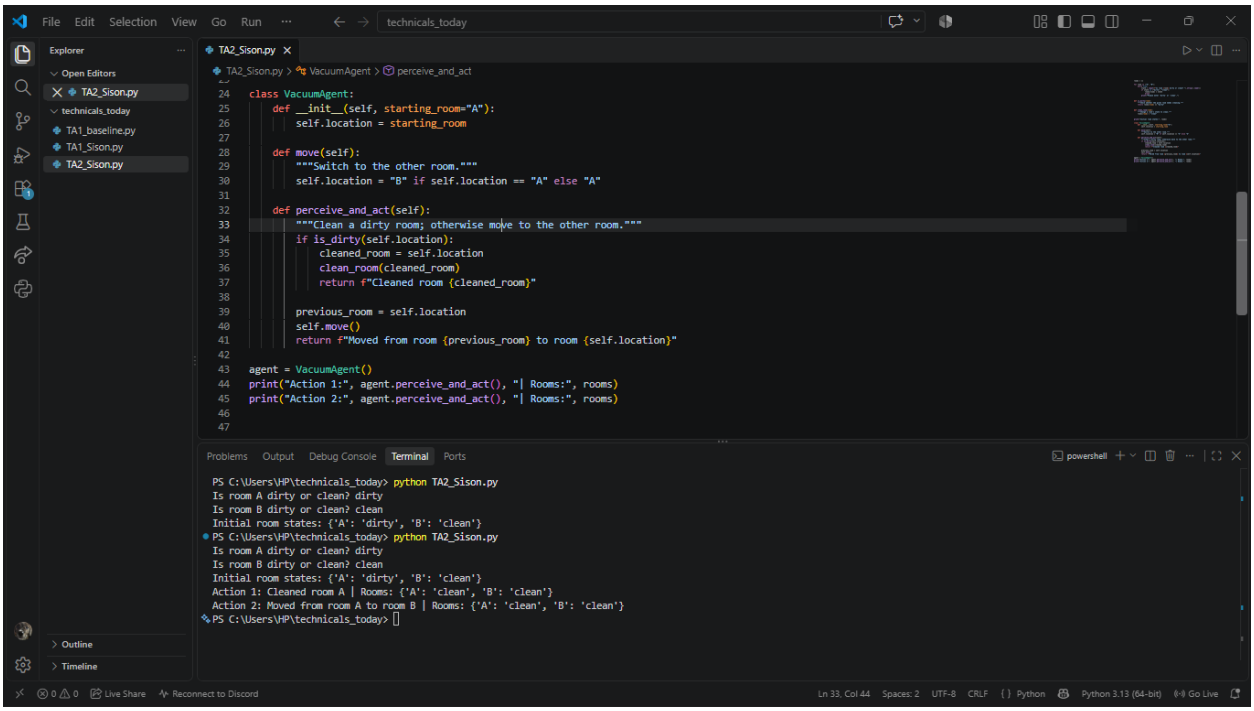
for room in ("A", "B"):
    while True:
        state = input(f"Is room {room} dirty or clean? ").strip().lower()
        if state in ("dirty", "clean"):
            rooms[room] = state
            break
        print("Please enter 'dirty' or 'clean'.")

def is_dirty(room):
    """Check whether the given room needs cleaning."""
    return rooms[room] == "dirty"

def clean_room(room):
    """Change a room's state to clean."""
    rooms[room] = "clean"

print("Initial room states:", rooms)
```

Task 2



Code:

```
rooms = {}

for room in ("A", "B"):
    while True:
        state = input(f"Is room {room} dirty or clean? ").strip().lower()
        if state in ("dirty", "clean"):
            rooms[room] = state
            break
        print("Please enter 'dirty' or 'clean'.")

def is_dirty(room):
    """Check whether the given room needs cleaning."""
    return rooms[room] == "dirty"

def clean_room(room):
    """Change a room's state to clean."""
    rooms[room] = "clean"

print("Initial room states:", rooms)

class VacuumAgent:
    def __init__(self, starting_room="A"):
```

```
        self.location = starting_room

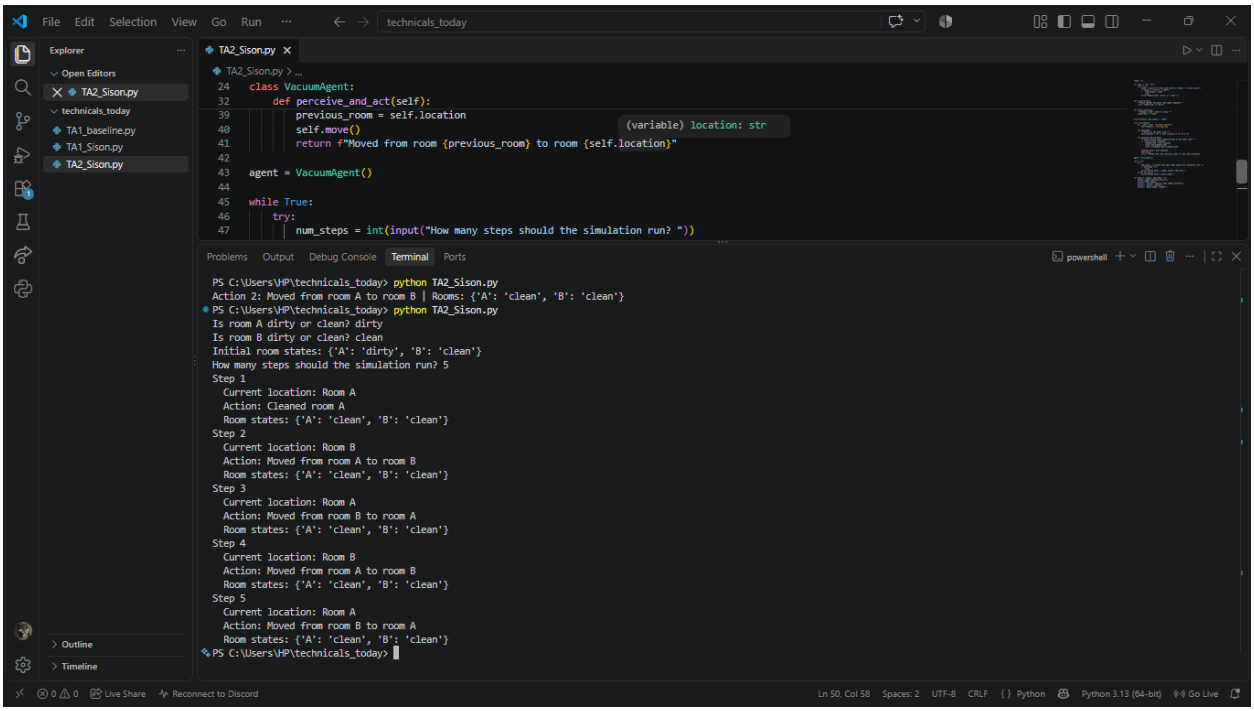
    def move(self):
        """Switch to the other room."""
        self.location = "B" if self.location == "A" else "A"

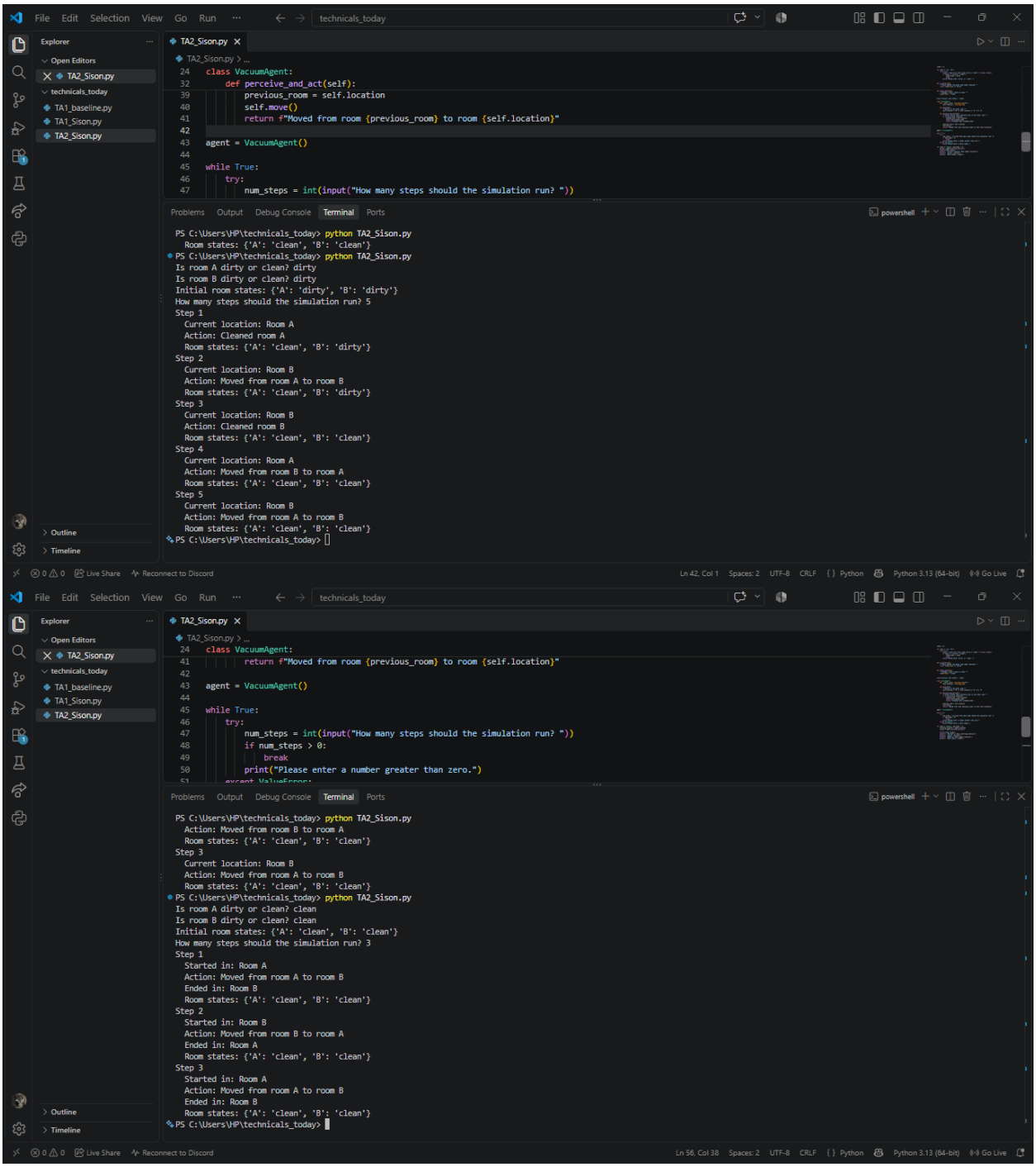
    def perceive_and_act(self):
        """Clean a dirty room; otherwise move to the other room."""
        if is_dirty(self.location):
            cleaned_room = self.location
            clean_room(cleaned_room)
            return f"Cleaned room {cleaned_room}"

        previous_room = self.location
        self.move()
        return f"Moved from room {previous_room} to room {self.location}"

agent = VacuumAgent()
print("Action 1:", agent.perceive_and_act(), "| Rooms:", rooms)
print("Action 2:", agent.perceive_and_act(), "| Rooms:", rooms)
```

Task 3




File Edit Selection View Go Run
technicals_today
Explorer
Open Editors
TA2_Sison.py
technicals_today
TA1_baseline.py
TA1_Sison.py
TA2_Sison.py
Outline
Timeline
TA2_Sison.py
TA2_Sison.py > ...
24 class VacuumAgent:
32 def perceive_and_act(self):
39 previous_room = self.location
40 self.move()
41 return f"Moved from room {previous_room} to room {self.location}"
42
43 agent = VacuumAgent()
44
45 while True:
46 try:
47 num_steps = int(input("How many steps should the simulation run? "))
Problems Output Debug Console Terminal Ports
powershell
PS C:\Users\HP\technicals_today> python TA2_Sison.py
Room states: {'A': 'clean', 'B': 'clean'}
PS C:\Users\HP\technicals_today> python TA2_Sison.py
Is room A dirty or clean? dirty
Is room B dirty or clean? dirty
Initial room states: {'A': 'dirty', 'B': 'dirty'}
How many steps should the simulation run? 5
Step 1
Current location: Room A
Action: Cleaned room A
Room states: {'A': 'clean', 'B': 'dirty'}
Step 2
Current location: Room B
Action: Moved from room A to room B
Room states: {'A': 'clean', 'B': 'dirty'}
Step 3
Current location: Room B
Action: Cleaned room B
Room states: {'A': 'clean', 'B': 'clean'}
Step 4
Current location: Room A
Action: Moved from room B to room A
Room states: {'A': 'clean', 'B': 'clean'}
Step 5
Current location: Room B
Action: Moved from room A to room B
Room states: {'A': 'clean', 'B': 'clean'}
PS C:\Users\HP\technicals_today>
Ln 42, Col 1 Spaces: 2 UTF-8 CRLF Python Python 3.13 (64-bit) Go Live
File Edit Selection View Go Run
technicals_today
Explorer
Open Editors
TA2_Sison.py
technicals_today
TA1_baseline.py
TA1_Sison.py
TA2_Sison.py
Outline
Timeline
TA2_Sison.py
TA2_Sison.py > ...
24 class VacuumAgent:
41 return f"Moved from room {previous_room} to room {self.location}"
42
43 agent = VacuumAgent()
44
45 while True:
46 try:
47 num_steps = int(input("How many steps should the simulation run? "))
48 if num_steps > 0:
49 break
50 print("Please enter a number greater than zero.")
Problems Output Debug Console Terminal Ports
powershell
PS C:\Users\HP\technicals_today> python TA2_Sison.py
Action: Moved from room B to room A
Room states: {'A': 'clean', 'B': 'clean'}
Step 3
Current location: Room B
Action: Moved from room A to room B
Room states: {'A': 'clean', 'B': 'clean'}
PS C:\Users\HP\technicals_today> python TA2_Sison.py
Is room A dirty or clean? clean
Is room B dirty or clean? clean
Initial room states: {'A': 'clean', 'B': 'clean'}
How many steps should the simulation run? 3
Step 1
Started in: Room A
Action: Moved from room A to room B
Ended in: Room B
Room states: {'A': 'clean', 'B': 'clean'}
Step 2
Started in: Room B
Action: Moved from room B to room A
Ended in: Room A
Room states: {'A': 'clean', 'B': 'clean'}
Step 3
Started in: Room A
Action: Moved from room A to room B
Ended in: Room B
Room states: {'A': 'clean', 'B': 'clean'}
PS C:\Users\HP\technicals_today>
Ln 56, Col 38 Spaces: 2 UTF-8 CRLF Python Python 3.13 (64-bit) Go Live

Code:

```
rooms = {}

for room in ("A", "B"):
    while True:
        state = input(f"Is room {room} dirty or clean? ").strip().lower()
        if state in ("dirty", "clean"):
            rooms[room] = state
            break
        print("Please enter 'dirty' or 'clean'.")

def is_dirty(room):
    """Check whether the given room needs cleaning."""
    return rooms[room] == "dirty"

def clean_room(room):
    """Change a room's state to clean."""
    rooms[room] = "clean"

print("Initial room states:", rooms)

class VacuumAgent:
    def __init__(self, starting_room="A"):
        self.location = starting_room

    def move(self):
        """Switch to the other room."""
        self.location = "B" if self.location == "A" else "A"

    def perceive_and_act(self):
        """Clean a dirty room; otherwise move to the other room."""
        if is_dirty(self.location):
            cleaned_room = self.location
            clean_room(cleaned_room)
            return f"Cleaned room {cleaned_room}"

        previous_room = self.location
        self.move()
        return f"Moved from room {previous_room} to room {self.location}"

agent = VacuumAgent()

while True:
    try:
```

```
            num_steps = int(input("How many steps should the simulation run? "))
            if num_steps > 0:
                break
            print("Please enter a number greater than zero.")
        except ValueError:
            print("Please enter a whole number.")

for step in range(1, num_steps + 1):
    starting_location = agent.location
    action = agent.perceive_and_act()

    print(f"Step {step}")
    print(f"  Started in: Room {starting_location}")
    print(f"  Action: {action}")
    print(f"  Ended in: Room {agent.location}")
    print(f"  Room states: {rooms}")
```

Task 4: Analysis Questions

After running the simulation, answer the following:

1. What rules does the agent follow?
2. How does the agent decide when to clean and when to move?
3. What happens if both rooms are already clean at the start?
4. How would you modify the agent to handle **more than two rooms**?

Answer the Questions ⇩

1. What rules does the agent follow?

- It follows two rules, this agent follows rule 1 if the current room is dirty to clean it, then rule 2 if the current room is clean which needs to move to the other room.

2. How does the agent decide when to clean and when to move?

- This agent bases its decision entirely on checking states of the location to see if its dirty or clean. If its dirty, it returns True for its current room where the action is to clean, if the dirty room returns false, the room is already clean, and the agent moves to another room.

3. What happens if both rooms are already clean at the start?

- The agent will just loop between rooms for the simulation as it'll never perform an action for cleaning.

4. How would you modify the agent to handle **more than two rooms**?

- I would definitely update the agent to handle more than two rooms like what's shown in the bonus task question, I'd use a list with room_names = ["A", "B", "C", "D"] then update the location by indexing with a modulo operator to cycle through all rooms in sequence.
- It would go something like what's shown below:

```
    def move(self):
        current_index = room_list.index(self.location)
        next_index = (current_index + 1) % len(room_list)
        self.location = room_list[next_index]
```